

The zero forcing number of $P(n, 3)$

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Abstract

In Theorem 3.6, Rashidi, Shajareh Poursalavati and Tavakkoli write that $Z(P(n, 3)) = 8$ for all n at least 12. Krishnan found that this statement is false for $n = 12$, where $Z(P(12, 3)) = 7$, calculated $Z(P(n, 3))$ exhaustively for n from 7 to 20, and conjectured that $Z(P(n, 3)) = 8$ for every n at least 13. He remarked that what was missing was a proof of a lower bound which holds for all large n . Here we provide that proof, and show that the conjecture holds. Our proof is a reduction. We observe that removing the interior coordinates of any matrix with the pattern of $P(n, 3)$ produces a single linear recurrence of order 8, so the nullity equals the dimension of the fixed space of the monodromy T , and is at most 8, with equality precisely when T is the identity. We then construct matrices achieving 8 by concatenating tiles, each having a fixed pattern for docking, so that their transfer products depend only on their interiors. There are tiles of every length from 21 up to but not including 42, and together they settle every n at least 21 at once. We certify each tile by evaluating a Krawczyk contraction in exact rational arithmetic. With per- n certificates for n from 17 to 20 and an exhaustive search for n from 13 to 16, we have proved $Z(P(n, 3)) = 8$ for every n at least 13. Here 13 is optimal, because $Z(P(11, 3)) = Z(P(12, 3)) = 7$.

1 Introduction

Let G be a graph whose vertices are each coloured black or white. The colour change rule is: if a black vertex has exactly one white neighbour, that neighbour becomes black. Let S be a set of vertices. Then S is zero forcing if colouring S black and then repeatedly applying the rule until it stalls colours all vertices. Let $Z(G)$ denote the smallest possible size of a zero forcing set. If $M(G)$ denotes the maximum nullity among all symmetric real matrices having exactly the edge set of G as their nonzero off-diagonal entries, then $M(G)$ is at most $Z(G)$ [1].

The generalized Petersen graph $P(n, k)$, for $n \geq 3$ and $1 \leq k < n/2$, has vertices $u_0, \dots, u_{n-1}$ and $v_0, \dots, v_{n-1}$, where u_i is joined to u_{i+1} , v_i is joined to v_{i+k} , and u_i is joined to v_i . All indices are taken modulo n . $P(n, k)$ is cubic.

In their Theorem 3.6, Rashidi, Shajareh Poursalavati and Tavakkoli [4] write that $Z(P(n, 3)) = 8$ for any $n \geq 12$. Krishnan [2] observed that this claim is false for $n = 12$, and gives an explicit set of seven elements which constitutes a zero forcing set for $P(12, 3)$. He also points out that an error was made in the proof of the result in the published paper, since it only rules out six-element sets in its case analysis. By brute force he determines $Z(P(n, 3))$ for $n = 7, \dots, 20$, finds that the values stabilise at 8 from $n = 13$ onward, and thus concludes:

Conjecture 5. $Z(P(n, 3)) = 8$ for any $n \geq 13$.

The obstacle is a matching lower bound valid for all n : the upper bound is available for every n at least 9, and only finitely many lower bounds can come from computation.

Theorem 1. $Z(P(n, 3)) = 8$ for every $n \geq 13$, and 13 is optimal, since $Z(P(11, 3)) = Z(P(12, 3)) = 7$. Moreover $Z(P(n, 3)) = M(P(n, 3)) = 8$ for every $n \geq 17$.

This is Conjecture 5 of Krishnan.

The upper bound $Z(P(n, 3)) \leq 8$ for n at least 9 is a rotation argument and is not new. The content is the lower bound, and all of it is in Sections 3–5. Section 3 reduces the kernel of any matrix carrying the pattern of $P(n, 3)$ to the periodic solutions of a linear recurrence of order 8, which bounds the nullity by 8 and identifies exactly when the bound is attained: when the monodromy of the recurrence is the identity. Section 4 makes that condition achievable for every large n simultaneously, by assembling the weight sequence out of tiles whose transfer products multiply independently. Section 5 certifies the tiles. Section 6 assembles the proof.

2 Setup

Fix n and k with $n \geq 2k + 3$. Let A be a real matrix indexed by the vertices of $P(n, k)$, with a nonzero entry in position xy if and only if xy is an edge, for $x \neq y$; the diagonal is unrestricted. We do not assume A is symmetric. Write

$$b_i = A_{u_i u_{i+1}}, \quad b'_i = A_{u_{i+1} u_i}, \quad c_i = A_{u_i v_i}, \quad c'_i = A_{v_i u_i}, \quad e_i = A_{v_i v_{i+k}}, \quad e'_i = A_{v_{i+k} v_i},$$

with diagonal entries a_i at u_i and d_i at v_i . All of b, b', c, c', e, e' are nonzero.

We do not need symmetry to prove that the nullity of A is at most $Z(G)$. We only need to know that there is a nonzero entry on every edge in both directions. If we have $Ax = 0$ and x is zero on a set B , and there is a vertex u in B such that u has exactly one neighbour w not in B , then all the terms in the row of $Ax = 0$ at u will be zero except the one involving x_w , which has a nonzero coefficient, so $x_w = 0$. We can show by induction over a forcing process that for any zero forcing set S , the restriction map $x \mapsto x|_S$ is injective on the kernel of A . So

$$\text{null } A \leq Z(G). \tag{1}$$

3 The reduction

This allows us to combine the outer and inner equations of the kernel of A into one recurrence along the outer cycle. Taking $k = 3$, we find that the recurrence is of order 8, which is the bound we want to match.

Theorem 2 (Reduction). *Let $n \geq 2k + 3$ and for $x \in \mathbb{R}^{\mathbb{Z}_n}$ define*

$$(Lx)_i = -\frac{b'_{i-1}x_{i-1} + a_i x_i + b_i x_{i+1}}{c_i},$$

obtained by solving the outer equation for the inner coordinate. Then the map taking x to the pair (x, Lx) is a linear isomorphism from

$$\left\{ x : \sum_{p=-k-1}^{k+1} \gamma_{i,p} x_{i+p} = 0 \text{ for all } i \in \mathbb{Z}_n \right\}$$

to the kernel of A , where $\gamma_{i,p} \neq 0$ only for $p \in \{-k-1, -k, -k+1\} \cup \{-1, 0, 1\} \cup \{k-1, k, k+1\}$, and the two extreme coefficients are

$$\gamma_{i,k+1} = -\frac{e_i b_{i+k}}{c_{i+k}} \neq 0, \quad \gamma_{i,-k-1} = -\frac{e'_{i-k} b'_{i-k-1}}{c_{i-k}} \neq 0.$$

Therefore, with T the monodromy of the recurrence on the state $(x_{i-k-1}, \dots, x_{i+k})$, the nullity of A is the dimension of the fixed space of T , is at most $2k + 2$, and equals $2k + 2$ if and only if T is the identity.

Proof. Every spoke weight is nonzero, so the equation of the kernel at u_i has a unique solution for the inner coordinate at i , a three-term expression in x . Putting this into the equation at v_i , namely $e'_{i-k}y_{i-k} + d_iy_i + e_iy_{i+k} + c'_ix_i = 0$, gives the formula: the three substituted terms give the three index triples $\{i - k - 1, i - k, i - k + 1\}$, $\{i - 1, i, i + 1\}$ and $\{i + k - 1, i + k, i + k + 1\}$, and the two extreme coefficients are as claimed, both nonzero since the outer, spoke and inner weights are. Every vector x solving all these equations gives a vector in the kernel.

Both extreme coefficients are nonzero, so the relation is a recurrence of order exactly $2k + 2$, which is solvable forward and backward, with solutions on the integers giving a space of dimension $2k + 2$, which the monodromy acts on, and whose n -periodic solutions are exactly the fixed vectors of the monodromy. This proves that the nullity of A is the dimension of that fixed space. This is at most $2k + 2$ and equals $2k + 2$ if and only if the monodromy is the identity. $\square$

This proof requires that n be at least $2k + 3$. Otherwise the index sets above collide modulo n , the extreme coefficients are no longer given by the explicit formulas, and for n equal to $2k + 1$ or $2k + 2$ the recurrence can have order less than $2k + 2$.

Corollary 3. $M(P(n, 3)) \leq 8$ for $n \geq 9$.

For $k = 3$ the order of the recurrence is 8, and the size of the forcing set produced by the rotation argument giving the upper bound is also 8. The upper bound and the ceiling of the matrix method are identical, with no slack: there exists a matrix on the pattern of $P(n, 3)$ with nullity 8 if and only if the monodromy of its reduced recurrence is the identity, and such a matrix proves the lower bound.

4 Tiles

The docking pattern consists of all weights equal to 1 in the outer, spoke and inner positions, and zero in the two diagonal positions. A tile of length ℓ is a group of ℓ consecutive positions in which the first $k + 1$ and last $k + 1$ positions have the docking pattern, and the other positions make up its interior. The tile product is the product of the transfer matrices over the positions of the tile.

Lemma 4 (Tile independence). *If the weight sequence is a concatenation of tiles sharing one docking pattern, each tile product depends only on that tile's interior weights, and the monodromy is the product of the tile products.*

Proof. By the reduction theorem, each transfer matrix depends on the coefficients of one reduced equation, involving weights only at indices in a window of width $2k + 2$ around its position. If the tile is in a block and the position is in that block, then the window will be in the block padded out by $k + 1$ on each side. The positions below the block will be in the last $k + 1$ positions of the previous tile and the positions above the block will be in the first $k + 1$ positions of the next tile; these also have the docking pattern, and so does the tile itself in its first and last $k + 1$ positions. Everything the transfer matrix sees is either docking or interior to this tile. The second statement says that the monodromy product associates when regrouped over the tiles. $\square$

Theorem 5 (Tiling). *Fix k and $L \geq 2(k+1)$. Suppose that for every length from L up to but not including $2L$ there are interior weights, all required entries nonzero, such that the tile product equals the identity. Then*

$$Z(P(n, k)) = M(P(n, k)) = 2k + 2 \quad \text{for every } n \geq L.$$

Proof. Each n at least L is a sum of integers from that range: if n is below $2L$ take the single term n , otherwise take a term L and recurse on $n - L$. We can concatenate together the corresponding tiles, giving a weight sequence with monodromy that by Lemma 4 is the product of the tile products, and hence the identity. All weights are nonzero, so it represents the pattern of $P(n, k)$, and from the reduction theorem the nullity is $2k + 2$. Hence $M(P(n, k)) \geq 2k + 2$; combining this with (1) and the rotation bound $Z \leq 2k + 2$ gives $M(P(n, k)) = Z(P(n, k)) = 2k + 2$. $\square$

We have proven this without gaps because we used a full interval. If the lengths of two tiles were coprime, then they could cover only nonnegative integer combinations of the two, and every n below a certain product would not be covered; for the smallest solvable lengths here, that uncovered range is already several hundred. Our hypothesis is a finite set of independent finite problems.

5 Certification

For $k = 3$ we take $L = 21$ and exhibit identity tiles of all lengths from 21 to 43 inclusive, which includes the needed range. That is 23 tiles.

To certify directly that the tile product is the identity would be to do interval arithmetic on a rational map. We do not need to do that. Applying the reduction theorem to the cyclic matrix formed from one tile tells us that the monodromy of that matrix is the tile product. Therefore the tile product is the identity if and only if the nullity of that matrix is $2k + 2$, and that last statement is about a bilinear system. Say K is a matrix with columns spanning a possible kernel, written in graph form. The property is that $A(w)K(X) = 0$, and its Jacobian is affine in the unknowns.

We take the square subsystem not because it is the maximum-rank selection one could make, but because it is chosen structurally. The product $K^T A K$ is symmetric, and its antisymmetric part accounts for precisely $\binom{8}{2} = 28$ dependencies among the equations, so removing those leaves a system equivalent to $AK = 0$, rather than a weaker consequence of it.

For each tile we then perform a Krawczyk contraction to certify it. With Y an approximate inverse of the square Jacobian J at a numerically located point, L_J a Lipschitz bound for J in the ∞ -norm, and δ a box radius,

$$\alpha = \|I - YJ\|_\infty + \|Y\|_\infty L_J \delta < 1 \quad \text{and} \quad \|Yf\|_\infty + \alpha\delta \leq \delta$$

together establish that a true real solution exists in that box.

Every number appearing here is evaluated exactly: the box centre is dyadic; the system is bilinear with integer coefficients, so the value and Jacobian at the centre are both exactly rational; and the approximate inverse is arbitrary, so rounding it to a rational costs nothing. Across all 23 tiles the contraction constant is at most 1.0×10^{-9} , and the nullity of each certified matrix is exactly 8. In addition, every docking weight is equal to its prescribed integer. Finally, none of the above uses any nullity calculation with floating-point numbers.

Performing exactly the same process per n rather than per tile gives us a proof that Z and M are both equal to 8 for every n between 17 and 33, that is, for seventeen consecutive numbers including the primes 17, 19, 23, 29 and 31.

6 Proof of the theorem

Proof of Theorem 1. First, the upper bound is correct for all $n \geq 9$, by the rotation argument.

For $n \geq 21$ we have $Z = M = 8$, since there are certified identity tiles of all lengths in the right range given by Section 5, so Theorem 5 applies with $L = 21$.

For n from 17 to 20: these are covered by the per- n certificates of Section 5, which give $Z = M = 8$ on the range 17 to 33.

Finally, for n in $[13, 16]$, exhaustive search on subsets of vertices shows there is no zero forcing set of size 7. Hence $Z = 8$ by the upper bound. We deduce Z only, and do not assert that $M = 8$ here.

Optimality of 13: exhaustive search gives $Z(P(11, 3)) = Z(P(12, 3)) = 7$. $\square$

Remark 6. The sequence of values for n from 7 to 14 is 6, 6, 6, 8, 7, 7, 8, 8. So $Z = 8$ first occurs at $n = 10$, well below the threshold 13 beyond which it always holds. These are two different quantities.

Remark 7. By the reduction theorem the lower bound is attained precisely when a monodromy equals the identity, a condition with no arithmetic in it. That is why the proof reaches all large n , including primes, and not a congruence class.

Independence and related work

This work was carried out independently. The lower bound for $n \geq 17$ was obtained on 3 September 2026 and the determination of $Z(P(n, 3))$ for every n on 22 September 2026; both dates are recorded in the version history of the project repository. I re-ran the verification suite for every computational claim in this note before posting.

After obtaining these results I learned of [3], a pull request to the repository `the-omega-institute/trureturing` (#10129, opened and merged on 26 September 2026), which states and claims a Lean proof of the same conjecture. I looked over it myself. What I verified is that it defines the statement

$$\text{claim} : \forall n, 13 \leq n \implies \text{zeroForcingNumber } (\text{gp } n \ 3) = 8,$$

which is Conjecture 5, that it contains a declaration `theorem result : claim`, and that its Lean sources contain no `sorry` and no additional `axiom` declarations. I did not build it, so I make no claim about whether it compiles, and I make no other assessment of it.

References

- [1] AIM Minimum Rank – Special Graphs Work Group, Zero forcing sets and the minimum rank of graphs, *Linear Algebra Appl.* **428** (2008), 1628–1648.
- [2] A. Krishnan, A correction to the zero forcing number of the generalized Petersen graphs $P(n, 3)$, arXiv:2607.19412 (2026).
- [3] `the-omega-institute/trureturing`, pull request #10129, “Prove Krishnan’s Conjecture 5: the zero forcing number of $P(n, 3)$ is 8 for every $n \geq 13$ ”, merged 26 September 2026. <https://github.com/the-omega-institute/trureturing/pull/10129>
- [4] S. Rashidi, N. Shajareh Poursalavati, M. Tavakkoli, Computing the zero forcing number for generalized Petersen graphs, *J. Algebra Combin. Discrete Struct. Appl.* **7**(2) (2020), 183–193.